Project Scope Management

Topic # 4B

Chapter 5 – Schwalbe

Chapter 5 – PMBOK

Requirement Gathering through Communications

- Interviews & Focus Groups
- Facilitated workshops
- User stories
- Questionnaires
- Observation
- Prototypes
- Benchmarks
- Existing System & Documents

Requirement Traceability Matrix (RTM)

- One method that can help track the requirements from the requirements document to the finished product is a *requirements traceability matrix (RTM).*

Requirements Traceability Matrix

Project Name:	
Cost Center:	
Project Description:	

ID	Associate ID	Requirements Description	Business Needs, Opportunities, Goals, Objectives	Project Objectives	WBS Deliverables	Product Design	Product Development	Test Cases
001	1.0							
	1.1							
	1.2							
	1.2.1							
002	2.0							
	2.1							
	2.1.1							
003	3.0							
	3.1							
	3.2							
004	4.0							
005	5.0							

Work Breakdown Structure (WBS)

- A WBS is a decomposition of the work required to complete the project.
- A WBS is made up of work tasks at the lowest level.
- It helps to improve the accuracy of time, cost & resource estimates & helps to communicate clear work responsibilities.
- The WBS is a hierarchical decomposition of the total scope of work to be carried out by the project team to accomplish the project objectives and create the required deliverables.
- The WBS organizes and defines the total scope of the project.

Creating WBS

- Decomposition is a technique used for dividing and subdividing the project scope and project deliverables into smaller, more manageable parts.
- The work package is the work defined at the lowest level of the WBS for which cost and duration can be estimated and managed.
- Expert judgment is often used to analyze the information needed to decompose the project deliverables down into smaller component parts in order to create an effective WBS.
- The project manager, in collaboration with the project team, then determines the final decomposition of the project scope into the discrete work packages that will be used to effectively manage the work of the project.

Why WBS?

- This helps you to objectively identify all of the work necessary without subconsciously leaving out real work in order to fit date constraints.
- The WBS step helps accomplish the following key objectives:
  - Develops an objective, rational view of the amount of work required
  - Helps team grasp the skills required and amount of resources required for the project
  - Provides a clear framework for assigning to individuals a clear task definition and delegate the responsibility for completion
  - Lays a foundation for analyzing the task dependencies and for isolating and managing risks
  - Lays a foundation for developing a bottom-up estimate for the project schedule
  - Allows tradeoffs to be made consciously and with the proper consensus involved.

Why WBS?

- **A WBS defines the required way to complete the project:**
- **A WBS creates a sense of urgency**
- **A WBS can prevent scope creep**
- **A WBS provides control**

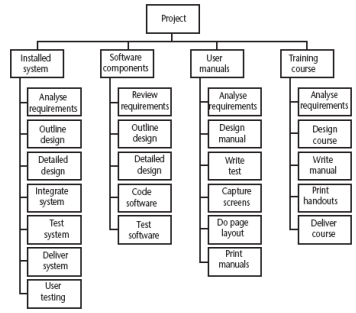
WBS Structure & Types

- WBS can have two forms:
  - Tabular Form
  - Like an Organizational Chart (Diagrammatic Form)
- WBS can be made through
  - Functions or Process
  - Phases
  - Deliverables
  - Cost Structures
  - Business Functions

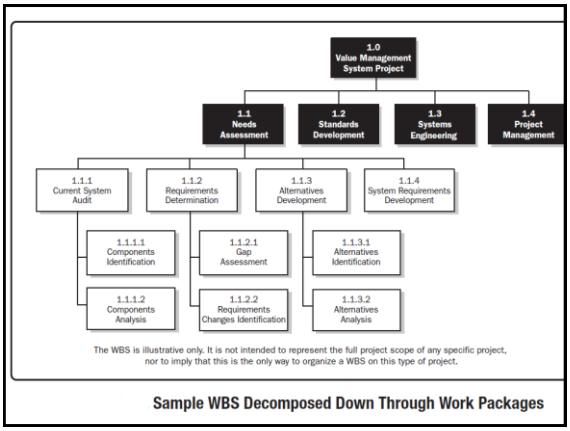
Identifying activities

- Work-based: draw-up a Work Breakdown Structure listing the work items needed
- Product-based approach
  - list the deliverable and intermediate products of project – product breakdown structure (PBS)
  - Identify the order in which products have to be created
  - work out the activities needed to create the products

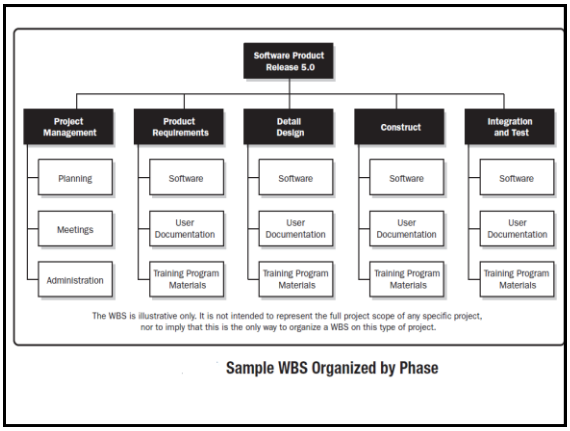
Hybrid approach



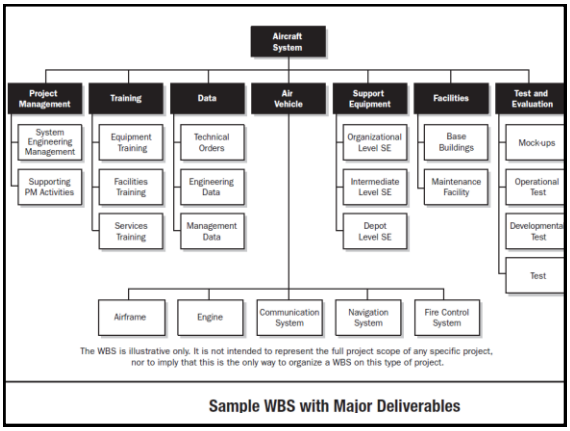
A Work Breakdown Structure based on deliverables



Sample WBS Decomposed Down Through Work Packages



Sample WBS Organized by Phase



Sample WBS with Major Deliverables

Example: WBS Organized by product

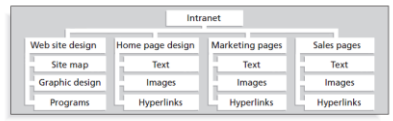
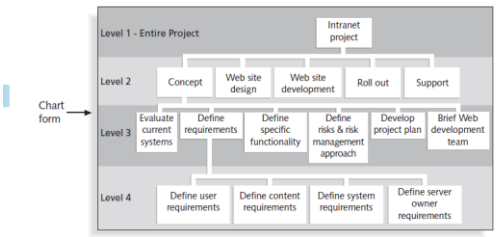


FIGURE 5-3 Sample intranet WBS organized by product

Example: WBS Organized by Phase



Example: WBS in tabular form

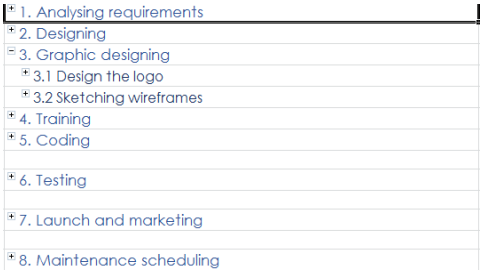
1.0 Concept
- 1.1 Evaluate current systems
- 1.2 Define requirements
  - 1.2.1 Define user requirements
  - 1.2.2 Define content requirements
  - 1.2.3 Define system requirements
  - 1.2.4 Define server owner requirements
- 1.3 Define specific functionality
- 1.4 Define risks and risk management approach
- 1.5 Develop project plan
- 1.6 Brief Web development team

2.0 Web site design
3.0 Web site development
4.0 Roll out
5.0 Support

WBS Construction

- Top Down v/s Bottom up
- **Top Down:**
  - Use deductive reasoning because it starts with general and is decomposed into more specific
  - Would identify a solution first & then dissect to implement the solution
  - Preferred method
- **Bottom-Up:**
  - Moves from more specific to general
  - Ideal for brainstorming session

Example - High level WBS of Market based Software Product



CASE STUDY

CASE ANALYSIS

WBS CONSTRUCTION OF A CAR POOLING APP

Scope Verification & Scope Change Control

- **Scope verification** involves formal acceptance of the completed project scope by the stakeholders.
  - This acceptance is often achieved by a customer inspection and then sign-off on key deliverables.
- **Scope control** involves controlling changes to the project scope.
  - Assure changes are processed according to procedures developed as part of integrated change control, and manage changes when they occur. You cannot do a good job of controlling scope
- A good job for collecting requirements, defining scope, and verifying scope is necessary for good control.

Managing & Controlling Scope Creep

- Develop a good project selection process for IT projects.
- Have users on the project team.
- Have regular meetings.
- Deliver something to project users & sponsors on regular basis.
- Co-locate users with the developers

Reducing Changing & Incomplete Requirements

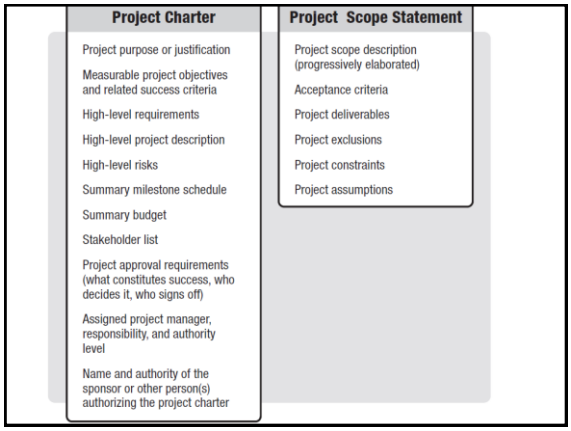
- Develop & Follow requirement management process *that includes procedure for initial requirements determination.
- Include techniques like prototyping, use case modeling & JAD to thoroughly understand user requirements.
- Put all requirements in writing & keep them current & readily available.
- Conduct testing & verification & validation throughout project life cycle.
- Use a process for reviewing requested requirements changes from a systems perspective.
- Emphasize completion dates.

Project Charter/Project Plan

- A **project charter** is a document that formally recognizes the existence of a project and provides direction on the project s objectives and management.
- **Components of Project Charter:**
  - The Project Title & Date of Authorization
  - Project Sponsor & Organization
  - Project Manager & Organization
  - Project Background
  - Goals & Objectives
  - Project Deliverables
  - Scope of project & exclusions
  - Risks & Assumptions
  - Critical Success Factors

Components of Project Charter

- Critical Success Factors
- Risk Mitigation & Contingency
- HL Plan & Milestones
- Cost estimates
- Stakeholders & responsibilities
- Project Structure & Organization Structure
- Job roles & Resource Requirements



END OF TOPIC 4B

-COMING UP!!!!!!
-Midterm Examination 1
-Team Analysis
-Project lifecycles
-Project Time Management

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